

PAUL PIOTROWSKI

Software Engineer · AI Agents & Full-Stack Web Applications

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EDUCATION

B.S. Computer Science — Florida International University

May 2025 – Dec 2027 | GPA: 3.80

Coursework: Data Structures, Algorithms, OOP, Java, C++, Discrete Structures, Statistics | Clubs: INIT, Google Developer Group

WORK EXPERIENCE

Software Engineer — Stride Rent a Car, Miami FL

Sep 2022 – Present

- Built and maintain a 6-module internal business dashboard covering Reservations, Claims, Reports, Chargebacks, Florida Renters, and Reviews. The dashboard integrates the Rently and HQ Car Rental APIs with a Supabase PostgreSQL backend and is used every day by the team to manage over 1,200 vehicles and roughly 5,000 reservations per month across a company of 100 employees, saving an estimated 45 hours of manual work per week.
- Cut chargeback response time by 90%, bringing the average from 30 minutes down to under 3 minutes. Instead of jumping between disconnected systems, the team now has all reservation, payment, and vehicle data in one place.
- Gave the team access to live fleet availability and reservation data for the first time, which improved pricing decisions, sped up customer responses, and reduced the amount of time vehicles sat idle between bookings.
- Serve as the sole engineer responsible for reservations, pricing, and chargeback workflows, taking projects from business requirement all the way to production for a company that processes over 5,000 reservations per month

Stack: React · TypeScript · Tailwind CSS · Supabase · Rently API · HQ Car Rental API · Vercel

Software Engineer Intern — Buo, Remote

Jan 2026 – Apr 2026

- Built the core dining hall data pipeline powering Buo's iOS app (live on the App Store), scraped real-time menu data from multiple college campus dining systems using Puppeteer and Playwright, enabling the app's flagship feature of personalized AI meal plans based on what's available in students' dining halls today.
- Designed and structured database tables to store and serve scraped menu data; integrated Mistral AI to normalize, classify, and enrich raw scraped meal data into structured nutritional records consumed by the app's meal planning engine.

Stack: JavaScript · Puppeteer · Playwright · Mistral AI · Database design

PROJECTS

PlateSpotter — AI License Plate Logging PWA

Feb 2026

Live: plate-spotter.vercel.app

- A production PWA used every day across 2 Stride offices, processing between 20 and 30 vehicle scans per day. It replaced a manual paper-based process and cut repair order logging time by 80%, bringing the average from 20 minutes down to under 4 minutes per order
- Architected a dual Gemini model pipeline. Pro Vision for AI camera scanning of license plates and Flash Lite for real-time NLP text formatting, enabling one-tap logging with no manual typing.
- Scan data syncs automatically to Google Sheets via a serverless Apps Script webhook across three log types (Repairs, Carwashes, and Returns), and the app works offline with full iOS and Android installability.
- Set up a GitHub Actions CI/CD pipeline for automated Vercel deployments on every push, ensuring the production app stays current without manual deployment steps.

Stack: TypeScript · React 19 · Tailwind CSS · Gemini Vision API · Vite · PWA · GitHub Actions · Vercel

ProfSummarizer — AI Lecture Tool

Jan 2026

Live: prof-summarizer.vercel.app

- Designed and deployed a full-stack AI application that transcribes audio lecture recordings and auto-generates Cornell Notes, flashcards, and configurable multiple-choice quizzes (5/10/15/20 questions) using Google Gemini 3.0 Flash.
- Architected a serverless backend on Supabase Edge Functions (Deno runtime) with independent service modules for transcription, summarization, and multi-turn chat, each independently scalable behind a clean REST interface.
- Currently used by 5 FIU students who collectively process around 50 lectures per week and generate 50 or more quizzes automatically. Users have reported an average 20% improvement in test scores after switching to AI-generated Cornell Notes and flashcard review
- Implemented Google OAuth authentication and Supabase row-level security so each user can only access their own notes and quiz history.

Stack: TypeScript · React 19 · Vite · Supabase (Edge Functions, PostgreSQL, Auth) · Gemini 2.0 Flash · GitHub Actions · Vercel

FinFlow — AI Personal Finance Dashboard

Jan 2026 · SwampHacks XI Hackathon

Live: finance-flow-lac.vercel.app

- Built at SwampHacks XI, integrated the Capital One Nessie API to fetch real-time bank transactions, automatically categorize spending, detect anomalies, and track savings goals across a full financial dashboard.
- Implemented a Gemini Pro conversational AI assistant with full financial context injected at runtime, enabling natural language Q&A over the user's own transaction history.

Stack: Next.js · TypeScript · MongoDB Atlas · Gemini Pro · Recharts · Framer Motion · Tailwind CSS

GradTrack — AI Academic Pathway Planner

Nov 2025 · Hackathon Project

- Built an AI-powered web app that helps students explore, plan, and visualize academic pathways and transfer options using real university course data, using Gemini function calling and search grounding to generate personalized program recommendations.
- Designed a modular TypeScript/React frontend with a service-based AI integration layer and secure API key management, keeping AI logic decoupled from UI components.

Stack: TypeScript · React · Google Gemini API (Function Calling, Search Grounding) · Tailwind CSS